Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Multimodal AI Platforms for Quantitative Cardiac MRI–Based Arrhythmia Risk Stratification			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	MRI	Medical Information & Systems	Other Medical Information & Systems
Concept of Proposed Technology	Current arrhythmia risk stratification for hypertrophic cardiomyopathy relies heavily on qualitative image interpretation and limited clinical variables, leading to uncertainty in intermediate-risk patients and inconsistent clinical decision making. This joint R&D program seeks proposals focused on AI systems that integrate quantitative tissue-level cardiac MRI biomarkers with longitudinal clinical data to enable: • Automated extraction of standardized quantitative imaging biomarkers (e.g., tissue characterization, structural remodeling, motion-based metrics); • Multimodal feature fusion combining imaging, clinical history, and temporal disease progression patterns; • Dynamic risk trajectory prediction supporting future risk estimation rather than single time-point classification; • Explainable and clinically interpretable outputs suitable for decision support.			

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	The program does not prescribe specific biomarkers, neural network architectures, or imaging vendors, emphasizing instead predictive performance, robustness, interpretability, and translational readiness.
Rationale for Joint R&D Project	Development of clinically reliable multimodal medical AI requires coordinated expertise in advanced imaging analytics, large-scale clinical data access, AI model development, software engineering, and regulatory-aligned system design. • U.S. research institutions provide access to clinically characterized cohorts for hypertrophic cardiomyopathy, disease-specific expertise, imaging physics knowledge, and translational validation capabilities. • Korean startups/SMEs contribute strengths in AI software engineering, system integration, quality management, and commercialization of medical software. A Korea–US joint R&D framework enables efficient transition from algorithm development to clinically deployable and globally scalable medical AI platforms.
Key Words	1. Artificial Intelligence (AI) 2. Cardiac MRI (CMR) 3. Arrhythmia Prediction (SCDA) 4. Quantitative Biomarkers 5. Multimodal Fusion

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	Proposals should aim to: 1. Develop automated quantitative cardiac imaging analysis pipelines; 2. Establish robust multimodal fusion and longitudinal risk modeling frameworks; 3. Demonstrate clinically meaningful predictive performance and interpretability; 4. Validate generalizability across institutions and imaging systems; 5. Prepare regulatory-aligned software architectures and clinical workflows.			
R&D Target Specifications	1. Predictive Performance: • AUROC: Internal > 0.88-0.92 • Balanced Accuracy: > 0.85 under event-balanced evaluation settings • AUPRC: >0.30 • Calibration (Brier score): < 0.13 2. Harmonization Performance: • Reduce inter-scanner intensity variability by > 40% 3. Quantitative Outputs: • Auto-extraction of LGE radiomics (> 200 features), T1/T2/ECV maps, and Strain (GLS/GCS/GRS) with variance < 5%.			
R&D Plan	**YEAR 1 — Data Integration and Algorithm Feasibility** Goal: Establish feasibility of quantitative imaging extraction and multimodal AI modeling. • Integrate cardiac MRI and clinical datasets for hypertrophic cardiomyopathy. • Develop automated quantitative imaging biomarker extraction pipelines. • Implement baseline multimodal fusion and preliminary risk prediction models. • Define evaluation metrics and interpretability strategies. • Conduct internal validation and feasibility assessment. Indicative Deliverables: • Functional quantitative imaging pipelines • Preliminary multimodal AI models • Initial performance and robustness evaluation **YEAR 2 — Model Optimization (TRL 4–6)** Goal: Demonstrate generalizable and clinically relevant predictive performance. • Optimize multimodal and longitudinal modeling architectures. • Implement data harmonization and domain adaptation strategies. • Develop prototype clinical decision-support interfaces. • Refine software architecture for scalability and security.			

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	Indicative Deliverables: • Optimized and validated AI models • Prototype clinical software system **YEAR 3 — Reliability, Regulatory Preparation, and Translation Readiness (TRL 6–7)** Goal: Establish regulatory readiness and commercialization pathways. • Demonstrate reproducibility and operational robustness of AI systems. • Finalize quality management and documentation frameworks. • Prepare regulatory-supporting technical files and validation evidence. • Conduct usability and workflow integration studies. • Establish commercialization and global deployment strategies. Indicative Deliverables: • Regulatory-ready software package • Clinical workflow validation results • Commercialization and partnership roadmap

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Imaging biomarker development and disease modeling • Clinical cohort curation and validation studies • Mechanistic interpretation and translational assessment
Expected R&R of Korean SME Partner	• AI model engineering and software system integration • Clinical workflow design and user interface development • Quality systems, regulatory preparation, and commercialization planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Improved precision of arrhythmia risk stratification • Reduced unnecessary invasive interventions and improved patient outcomes • Globally scalable quantitative imaging AI platforms • Strengthened Korea–US collaboration in medical AI innovation

6. References

1. Nat Cardiovasc Res. 2025 Jul;4(7):891-903. doi: 10.1038/s44161-025-00679-1.
2. JACC Cardiovasc Imaging. 2024 Aug;17(8):880-893. doi: 10.1016/j.jcmg.2024.04.013. Epub 2024 Jul 10.PMID: 39001729
3. JACC Adv. 2025 Nov;4(11 Pt 2):102307. doi: 10.1016/j.jacadv.2025.102307.